LiDAR Car

Meeting 1 - 2.9.2026

Welcome !

- I’m Pranav (sophomore in ece)
- I like being outside, baking, making/playing games
- Quickly go around and share name, year, major, and how many grapes would you eat if each one was $10k but one was poisoned

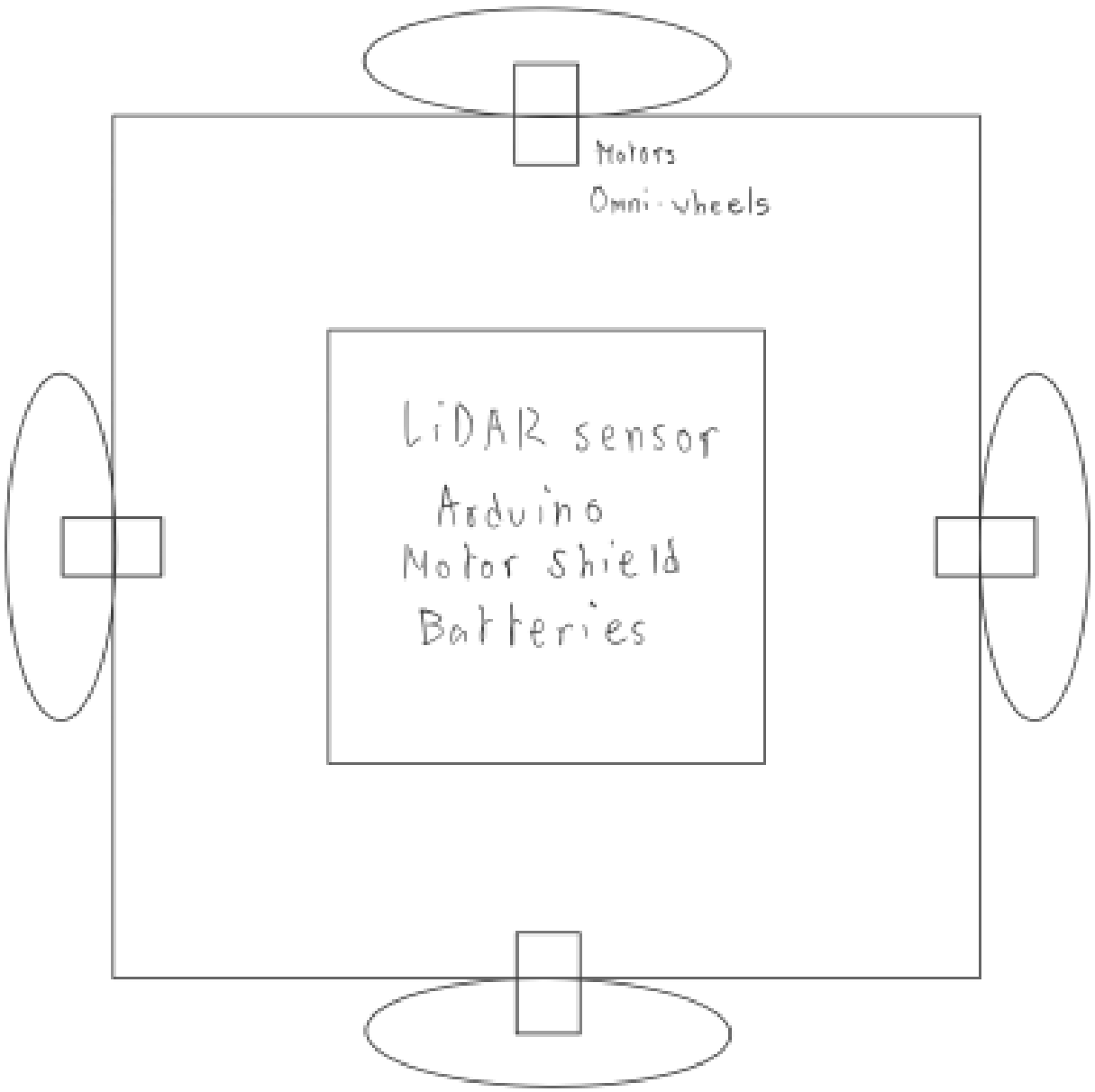
Project overview

- We are going to build an autonomous robot that can navigate a maze using a LiDAR sensor and FPGA control.
- We need to process 7,000+ sensor measurements per second with the FPGA, write an algorithm that makes smart and quick navigation decisions, then translate those decisions into precise motor movements to move the robot through the maze.

*(Please ask questions at any point)

The Robot

- Square body, 3d printed chassis
- Omnidirectional wheels (no turning)



Technical overview

2D LiDAR Scanner
7,000 samples/sec

← Senses environment

UART, Raw sensor data

FPGA
- Parse sensor data
- Find open space
- Choose direction

← Processes data + decides direction

UART, Motion command

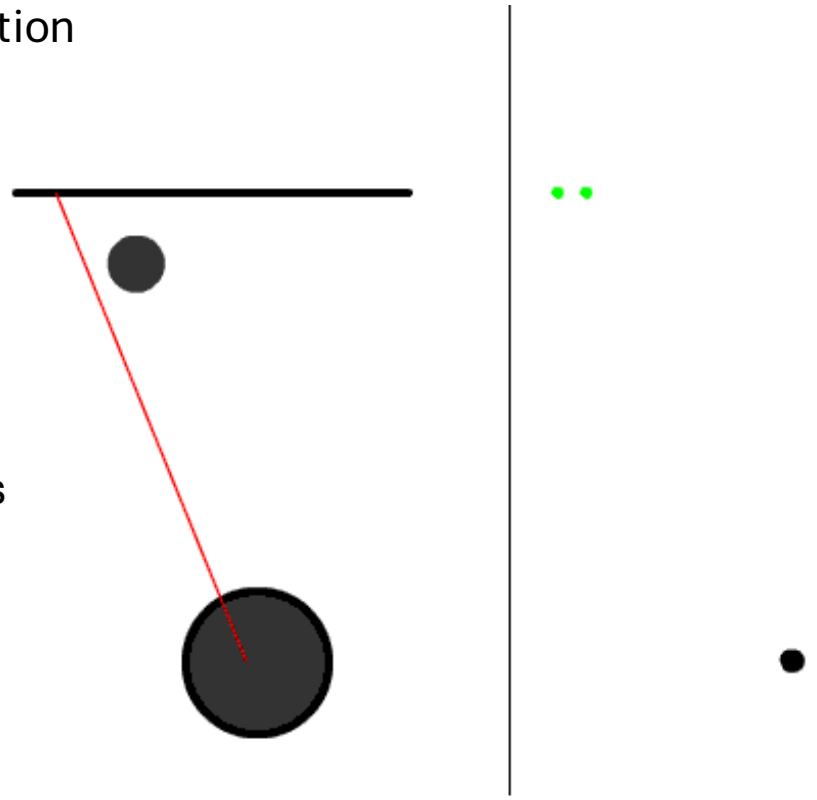
Arduino + Motor.
Shield

← Translates to wheel movements

Motor control

4 Independent Motors
+ Omni Wheels

← Robot moves through maze



Team Breakdown

- Software team
  - Work with LiDAR sensor (UART communication, packet parsing)
  - Write SystemVerilog code for FPGA –
  - Design the navigation algorithm
  - Debug with logic analyzers and simulation tools
- Hardware team
  - Program Arduino for motor control
  - Build power system (batteries, voltage regulation)
  - Figure out kinematics for wheels (direction → wheel speeds)
  - Design and 3D print robot chassis

No meeting next week ! (presidents day)

- Our next meeting will be on Feb 23
- Think about what team you want to be on
- We will be working with components next meeting
  - Software team: write C++ code to read LiDAR data
  - Hardware team: learn motor control via Arduino
- Please react to Sam's message in the LiDAR maze channel on discord to get a role

- **Bring cardboard to meetings!!!** We will need it to build our maze. White/lighter colors work better (light reflects better)

Any questions?